

MRI Image Quality Metrics (IQMs)

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Abstract

This document summarizes commonly used MRI image quality metrics (IQMs) and their mathematical definitions for reproducibility and reporting.

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Introduction

This document summarizes commonly used MRI image quality metrics (IQMs) and their mathematical definitions for reproducibility and reporting.

1 Per-Slice IQMs (Foreground / Background pixels)

All metrics below accept foreground pixels $\mathbf{F}$ (from segmentation) and background pixels $\mathbf{B}$; NaN/Inf inputs yield NaN. When a denominator is zero or data are missing, the metric is NaN by construction.

1.1 Mean (MEAN)

Foreground mean intensity:

$$\text{MEAN} = \mu_F = \frac{1}{N_F} \sum_i F_i$$

1.2 Range (RNG)

Foreground intensity range:

$$\text{RNG} = \max(F) - \min(F)$$

1.3 Variance (VAR)

Foreground variance:

$$\text{VAR} = \sigma_F^2 = \frac{1}{N_F} \sum_i (F_i - \mu_F)^2$$

1.4 Coefficient of Variation (CV)

Foreground coefficient of variation (percent):

$$\text{CV} = \frac{\sigma_F}{\mu_F} \times 100$$

1.5 Contrast per Pixel (CPP)

Laplacian-filtered contrast, mean response of a 3×3 Laplacian:

$$f_1 = \frac{1}{8} \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}, \quad \text{CPP} = \text{mean}(f_1 * F)$$

where $*$ is 2D convolution (foreground is median-filtered only when used for PSNR below).

1.6 Peak Signal-to-Noise Ratio (PSNR)

$$\text{PSNR} = 10 \log_{10} \left(\frac{\max(F)^2}{\text{MSE}(F, \hat{F})} \right)$$

where $\hat{F}$ is either the background $\mathbf{B}$ (if provided) or a median-filtered version of $\mathbf{F}$.

1.7 SNR1

Ratio of foreground to background standard deviation:

$$\text{SNR1} = \frac{\sigma_F}{\sigma_B}$$

1.8 SNR2

Foreground mean over background standard deviation:

$$\text{SNR2} = \frac{\mu_F}{\sigma_B}$$

1.9 SNR3

Foreground standard deviation divided by centered-foreground standard deviation (stability check):

$$\text{SNR3} = \frac{\sigma_F}{\sigma_{F-\mu_F}}$$

1.10 SNR4

Foreground mean over background mean:

$$\text{SNR4} = \frac{\mu_F}{\mu_B}$$

1.11 Contrast-to-Noise Ratio (CNR)

Foreground/background mean difference over background standard deviation:

$$\text{CNR} = \frac{\mu_F - \mu_B}{\sigma_B}$$

1.12 Patch Coefficient of Variation (CVP)

Foreground patch coefficient of variation:

$$\text{CVP} = \frac{\sigma_F}{\mu_F}$$

1.13 Coefficient of Joint Variation (CJV)

Joint variation between foreground and background:

$$\text{CJV} = \frac{\sigma_F + \sigma_B}{|\mu_F - \mu_B|}$$

1.14 Entropy Focus Criterion (EFC)

Normalized Shannon entropy of foreground:

$$p_i = \frac{F_i}{\sqrt{\sum_j F_j^2}}, \quad \text{EFC} = -\frac{1}{\log N_F} \sum_i p_i \log p_i$$

higher values suggest more ghosting/noise.

1.15 Foreground-Background Energy Ratio (FBER)

Median energy ratio:

$$\text{FBER} = \frac{\text{median}(F^2)}{\text{median}(B^2)}$$